
| RESEARCH ARTICLE

Data-Driven Healthcare Operations and Patient-Centric Quality Improvement: Integrating Sentiment Analytics, Intelligent Scheduling, and Queueing Theory

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| ABSTRACT

As patient experience continues to be recognized and gain traction within a data-driven healthcare operation, the world of quantitative patient experience is expanding, specifically to improve the quality, efficiency, safety, and access to healthcare for patients. This conceptual study suggests a patient-centric framework that connects sentiment analytics and a theory of intelligent scheduling and queueing together to link patient sentiment to operational decision-making. Sentiment Analytics can be used to gain insight into the emotional content of patient stories, reviews, surveys, and beyond to reveal emotional responses, service concerns, and specific areas for improvement. Intelligent scheduling helps coordinate patients, staff, resources, patient preferences, and clinical workflows, and queueing theory offers quantitative tools to understand the patient flow, waiting times, capacity, utilization, and service priorities. Furthermore, the framework also covers aspects of data quality management, interoperability, privacy by design, minimum necessary access, ethical data governance, and continuous data validation to assist in the implementation of it reliably and responsibly. If patient feedback systems, scheduling, and capacity interventions are integrated with EHRs, there can be a continuous improvement loop. Patient experiences can initiate problem identification, models quantify operational conditions, and the effects of interventions are evaluated according to the results and feedback. The use of patient experience data in tandem with operational data analysis offers a systematic framework for enhancing patient flow, minimizing delays, maximizing the use of resources, improving patient engagement, and a more patient-centric, responsive, and equitable approach to healthcare delivery.

| KEYWORDS

Sentiment Analytics, Intelligent Scheduling, Queueing Theory, Patient-Centric Care, Healthcare Quality Improvement

| ARTICLE INFORMATION

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1. Introduction

Healthcare delivery is now a complex, data-driven, patient-centric system, where addressable clinical quality, operational efficiency, safety, and patient experience are concurrent. While healthcare has always been a fundamental part of human existence, healthcare systems today are delivered via a variety of delivery points, through a wide range of multidisciplinary teams, supported by sophisticated information technologies, and under ever more complex patterns of patient demand [Han, 2024]. As a result, conventional methods, which rely mainly on the retrospective analysis of performance, are increasingly supplemented by methods that are based on data. Therefore, there is a growing need to combine conventional methods, which are mainly based on the retrospective analysis of performance, with data-driven methods that can identify patient needs, operational bottlenecks, and

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quality gaps in real time. In this broader context, sentiment analytics, intelligent scheduling, and the theory of queues are just ideas for using patient experience and operational data to drive action steps that can ultimately lead to better healthcare outcomes [Akinode, 2017]. The proposed framework combines three complementary methodological pillars: sentiment analytics, intelligent scheduling, and queueing theory. Sentiment Analytics can convert qualitative patient comments, reviews, complaints, and other unstructured text into quantifiable data of patient perceptions and experiences. From lexicon-based methods to traditional machine learning, deep learning, natural language processing, and large language models, increasingly large and diverse sets of patient narratives have been studied [Niu, 2024]. These methods may help to uncover positive and/or negative experiences, common issues, communication difficulties, and service trends that may not be easily identified in traditional structured surveys. The information generated can be used in quality improvement efforts and can also offer an extra patient-centered aspect to operational decision-making. Intelligent scheduling complements sentiment analytics by linking the needs and preferences of patients with scheduling appointments, staff, facility, and other healthcare resources [Ridoy, 2022]. The timing of appointments directly affects access to care, workload distribution, patient flow, and waiting times. With operational data and patient feedback, scheduling processes can be adjusted through potential increases in responsiveness to actual demand and patient experience. This integration enables healthcare providers to take into account not only appointment frequency and timing, but also issues like the perception of delays, communication problems, service priorities, and repeat areas of dissatisfaction [Kammrath, 2025]. This allows scheduling to be structured as a means of service delivery that will help realize patient-centered quality goals. The quantitative basis of the understanding of patient flow, waiting time, capacity utilization, and service levels is given by the queueing theory. In healthcare environments, the demand for services is variable, service times are unpredictable, resources are limited, and services are multifaceted with multiple back-to-back stages. These attributes are operational challenges in certain areas such as EDs and other areas of high demand, where waiting and congestion are issues [Li, 2021]. In queueing theory, there are mathematical models that can be used to study how arrival rates relate to service capacity, utilization, and waiting times. They can be integrated with scheduling and patient experience data to make better staffing, appointment capacity, resource allocation, and service redesign decisions. Data-Driven Healthcare Operations and Patient-Centric Quality Improvement is a conceptual merging of qualitative patient experience data and quantitative operational analytics [Knight, 2023]. The approach aims to bridge the gap between the patient experience and the organisation and delivery of healthcare services by linking what patients experience to the current quality improvement framework and healthcare governance. The core thesis is that the quality of healthcare can be improved when patients' perspectives, operational performance, and clinical goals are analyzed simultaneously [Saad, 2023]. This integration could offer healthcare institutions a framework for enhancing access, minimizing avoidable wait times, optimizing resources, optimizing patient experiences, and enabling better care delivery and responsiveness.

2. Sentiment Analytics and Intelligent Scheduling for Patient-Centric Healthcare Improvement

Sentiment analytics and intelligent scheduling are just two aspects of data-driven improvement in healthcare. Sentiment analytics extracts structured data from qualitative patient feedback to uncover perceptions, concerns, and emotional reactions related to the healthcare experience, and intelligent scheduling correlates patient demand with clinical processes, resources, and capacity [Ang, 2022]. Combining these methods can help healthcare providers go beyond satisfaction and appointment attendance rates to gain a deeper understanding of the patient journey and the impact of their processes on the patient experience. Healthcare sentiment analytics uses natural language processing (NLP) and other AI methods to detect and classify emotions, experiences, opinions, and concerns from various patient data sources (Table 1). It helps health care providers gain insight into patient perceptions, the gaps in care, and the interventions that can be made at an opportune time to enhance satisfaction and outcomes [Zhou, 2021]. In addition to detecting positive or negative sentiment, advanced sentiment analytics can detect more subtle emotions like happiness, sadness, anger, fear, frustration, or surprise. This data can be especially useful when used to gain insight into patient experiences throughout the entire healthcare process and when understanding problems that may not be revealed by traditional numerical measures of patient satisfaction. Sentiment analytics in healthcare can pull various data from patient reviews, surveys, social media posts, electronic health records, chatbot interactions, telephone conversations, or call center transcripts [Caldwell, 2024]. Another potential clinical application is to examine clinical notes, discharge summaries, and other provider-patient

interactions for trends related to treatment experiences, communication quality, and provider interactions with healthcare teams. Data should be gathered and analyzed within the confines of the applicable ethical and regulatory guidelines. Sentiment analysis and intelligent scheduling are complementary approaches for enhancing patient-centric healthcare operations [Zhang, 2026]. Sentiment Analytics tells you how patients feel about a healthcare service, while intelligent scheduling tells you how to organize a healthcare service more effectively. They can offer a systematic framework for healthcare organizations to use for recognizing service issues, optimizing resource use, minimizing avoidable delays, increasing patient involvement, and enhancing the quality of care.

Table 1. Integrated Analytical Approaches for Patient-Centric Healthcare Improvement

Analytical component	Primary function	Healthcare application	Expected contribution to quality improvement	Reference
Sentiment analytics	Extracts emotions, opinions, concerns, and experiences from unstructured patient feedback	Patient reviews, surveys, EHR narratives, chatbots, call transcripts	Identifies patient concerns, communication problems, satisfaction patterns, and service weaknesses	[5], [27], [31]
Aspect-based sentiment analysis	Links sentiment to specific aspects of healthcare services	Nursing care, waiting time, cleanliness, communication, accessibility	Enables targeted interventions rather than relying only on overall satisfaction scores	[31]
Intelligent scheduling	Coordinates appointments, staff, facilities, and patient demand	Outpatient services, clinics, diagnostic services	Reduces scheduling conflicts, no-shows, workload imbalance, and unnecessary delays	[36], [37], [38]
Queueing theory	Models arrival rates, service capacity, utilization, and waiting times	Emergency departments, outpatient clinics, inpatient services	Supports capacity planning, patient flow optimization, staffing decisions, and waiting time reduction	[9], [10]
Discrete event simulation	Reproduces complex healthcare processes under different operating conditions	Emergency departments, vaccination clinics, outpatient pathways	Enables testing of operational interventions before real-world implementation	[9], [10]

3. Queueing Theory in Healthcare Service Systems

Queueing theory is a valuable analytical tool for understanding patient flow, delays, resource capacity, and patient allocation in healthcare service systems. The complexity of queues in healthcare facilities often lies in the variability of patient arrivals, services provided, service times, staffing levels, and demands for services, especially within emergency departments, outpatient clinics, diagnostic departments, and inpatient departments [Gao, 2022]. Mathematical queueing models offer a systematic approach to capturing these processes and analysing the impact of operational policies on waiting times, service utilisation, congestion, and system performance. Queueing theory, when used in conjunction with data-driven analytics and simulation, can help improve healthcare efficiency and patient experience. A major use of queueing theory is in the management of EDs. Emergency departments are required to treat patients with varying priorities and face unpredictable patient flow and limited resources [Valenzuela-Núñez, 2024]. Queueing models can be used to model the flow of patients through various phases of an emergency service and to analyse the impact of different patient flow arrangements. For instance, patients may be divided into different fast track pathways according to clinical need, expected complexity of treatment or acuity. These pathways can be simulated to assess possible variations in waiting times, use of resources and patient flow before implementing operational changes in real life. Another important application of priority queueing in the health care sector is the selection of patients for dialysis. Patients do not have the same clinical needs or urgency levels, and in clinical settings a first-come, first-served approach is not appropriate [AISerkal, 2025]. Priority-based

models can be used to rank patients for the services based on any clinically relevant variable. These models can be applied to analyze the effects of priority rules on waiting times between different patient groups and the allocation of scarce resources under different priorities while keeping waiting times at an acceptable level. This is especially important in the ED, where there are high acuity patients who need to be seen quickly, and low acuity patients who still need to receive their appropriate services [Yang, 2024]. Queueing theory is a valuable tool for the study of healthcare service systems, as it links the demand for services, the capacity of the system, the waiting times for patients, the utilization of resources, and the policies governing the operation of the system. It can be used in conjunction with queueing networks, priority models, discrete event simulation, and data-driven analytics to analyse complex patient flow problems in multiple ways [Haleem, 2021]. With proper validation, context, and interpretation, queueing-based approaches can enable the improvement of staffing decisions, routing of patients more efficiently, shorter waiting times, better utilization of capacity, and better responsiveness of the healthcare services. Its greatest value is therefore in the combination of mathematical modelling with empirical data, managerial experience, and patient-based quality improvement goals.

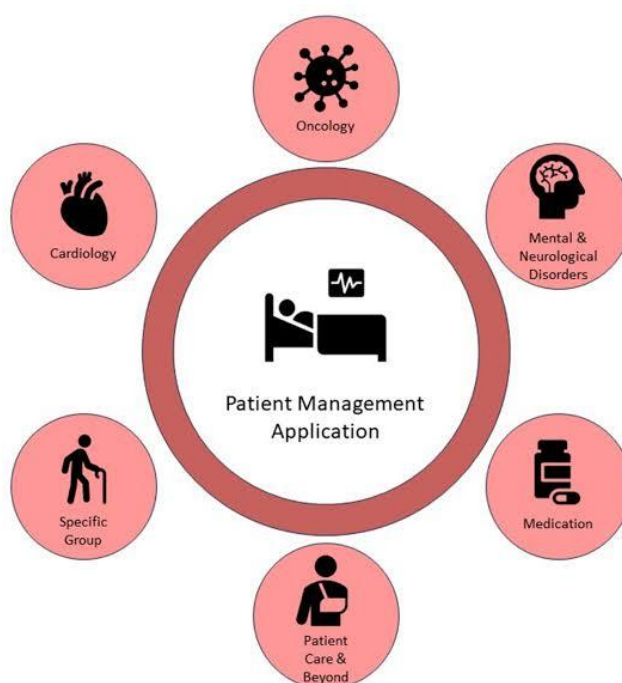


Figure 1. Data-driven decision making in patient management [Laan, 2018]

4. Data-Driven Patient-Centric Quality Improvement and Integrated Healthcare Operations

A data-driven patient-centric quality improvement framework combines high-quality healthcare data, patient feedback, and analysis to inform ongoing improvement of healthcare operations. Access to accurate, reliable, and deduplicated patient data is vital across healthcare practices and across EHR systems to enable the proper measurement of quality, evaluation of patient outcomes, and determination of whether interventions have a meaningful impact [Yadav, 2023]. Good data stewardship is, therefore, the bedrock of trustworthy patient experience, operational performance, and clinical outcome measurement. The relationship between healthcare structure, processes, outcomes, and patient experience could form the basis for quality measurement in such a framework. Care processes affect the quality and efficiency of the interactions between the patient and the healthcare provider, and organizational resources, infrastructure, staffing, and governance affect how care is delivered. The outcomes that result indicate whether or not health goals have been met. Patient experience should be integrated across this measurement structure as it offers firsthand data on accessibility, communication, timeliness, responsiveness, and perceived quality of care [Obermeier, 2020]. This means accurate, complete, accessible, and usable information in the EHR is vital to meaningful quality assessment activity throughout the care cycle. The framework can also be linked with the significant dimensions of quality of health care such as safety,

effectiveness, patient centeredness, timeliness, efficiency, and equity. Combining these dimensions with patient-reported indicators gives a more comprehensive picture, as all outcomes, performance, patient experience, and patient safety dimensions are taken into account [Yadav, 2019]. This method enables priority setting of improvement initiatives based on system performance measures as well as patient needs along the care journey. The framework brings together the perspectives of patients with measurable operational conditions and provides mechanisms to turn those comments into actionable interventions.

5. Data Quality, Ethical Governance, and Implementation in Healthcare Organizations

Data quality management is a core component of the data-driven, patient-centred approach to healthcare improvement, as good decisions rely upon accurate, consistent, complete, and usable information from across a range of sources. These sources vary in governance, terminology, completeness, and collection processes, so it is important to have consistent governance, validation, monitoring, and staff training, which is essential to ensure a consistent level of data quality. High-quality data can be used to represent patient experiences as a general sentiment category as well as using more granular measures, which can lead to a more accurate assessment of patient satisfaction, concerns, and service quality [Alam, 2023]. The success of integration across diverse sources depends on ensuring consistent data formats, validation methods, and monitoring efforts. Data schemas should be designed in a way that allows for the seamless integration of data from various sources, such as reviews, clinical records, automated conversations, and operational systems, without causing any inconsistencies or duplication (Table 2). The application of privacy-preserving techniques should also be uniform across sources, so that integrated data sources maintain their analytical utility while maintaining the privacy of patient information [Shama, 2023]. Another aspect of cross-source integration that needs to be taken into account is the generalizability of analytical models. The performance of a model developed from the data from one setting, population, source, or time period can vary when applied to new data from the real world. It is therefore important to independently validate analytical output with diverse and representative datasets to see if analytical outputs remain reliable when data sources and health care environments change. A robust healthcare improvement system relies on the synergy of trusted data, effective multi-source linkage, privacy safeguards, ethical governance, and organizational action [Adhicandra, 2024]. Data quality ensures that the outputs of the analytics are reliable and credible; linking controls help to interpret information from multiple sources together; privacy mechanisms ensure that sensitive information is protected; and governance structures ensure accountability (Figure 2). Together with patient feedback, sentiment analytics, intelligent scheduling and operational analysis through queueing, a continuous improvement process can be built that links patient experiences with tangible operational improvements [Yousif, 2022]. This will enable better decisions to be made, whilst at the same time providing privacy for patients, regulatory compliance, engagement among staff and quality improvement from a patient-centred perspective.

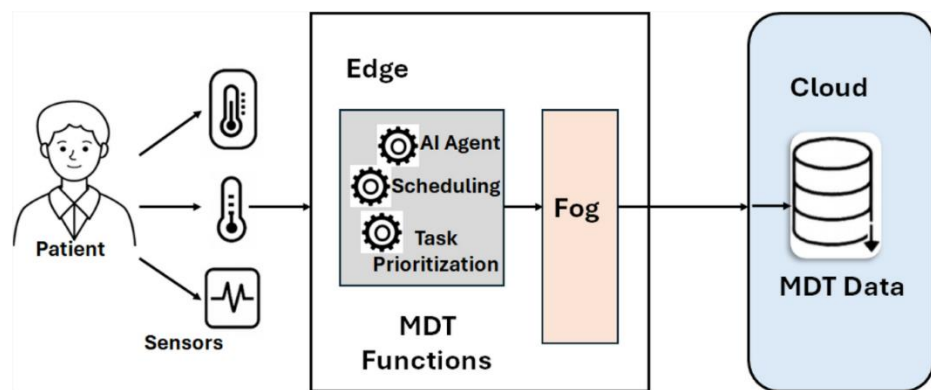


Figure 2. A cost-optimized medical digital twin framework for secure and efficient patient data management in smart healthcare [Zuo, 2024]

Table 2. Data Governance, Quality, and Ethical Requirements for Implementation

Domain	Key requirement	Implementation approach	Importance for healthcare analytics	Reference
Data quality	Accuracy, completeness, consistency, and standardization	Validation, data cleaning, standardized formats, monitoring, and staff training	Improves reliability of sentiment analysis and operational decisions	[41], [43]
Cross-source integration	Consistent linking of heterogeneous datasets	Schema alignment, standardized identifiers, validation, and interoperability	Enables reliable integration of patient, clinical, and operational information	[43], [44]
External validation	Assessment of model generalizability	Independent and diverse datasets, human benchmarking, and continuous testing	Reduces risks associated with poor performance in new populations or settings	[5]
Privacy protection	Safeguarding sensitive patient information	De-identification, anonymization, encryption, and controlled access	Reduces privacy risks while preserving analytical utility	[27], [44]
Minimum necessary access	Limiting protected health information to what is required	Role-based access, redaction, release controls, and authorization procedures	Reduces unnecessary disclosure of protected health information	[16], [44]

6. Discussion

Data-driven analytics and patient-centred quality improvement create a complete solution to tackle the clinical and operational issues that healthcare organizations are dealing with together. The patient insights, sentiment analysis, intelligent scheduling, queueing theory, data quality management, and governance are integrated in a single improvement structure that this study has developed [Hur, 2026]. The main concept in the conceptual framework is that patient experience is not just a satisfaction score. Rather, it should link directly to the operation (i.e., availability for appointments, staffing, patient flow, waiting times, and resource utilization). This holistic view offers a chance to shift the focus of problem identification from the past to a present and evidence-based approach to improving healthcare services. Patient feedback is a very important aspect as it helps to gather feedback that cannot be gathered through any other conventional clinical and operational measures. Healthcare is experienced from multiple angles by patients, which include communication, accessibility, timeliness, responsiveness, continuity, and interactions with healthcare professionals [Rahman, 2026]. These experiences may have an impact on perceptions of quality even if the clinical outcomes are satisfactory. In the framework, therefore, patient feedback is placed as a strategic source of information to find the weak points in services and to prioritize the activities for service improvement. Feedback that is regularly gathered, analysed and tied to action plans can enable learning within an organisation and redesign of services. This also helps cultivate a non-punitive quality improvement culture and a culture of feedback that focuses on the weaknesses of the system and not on the individual [Olayode, 2021]. Sentiment analysis improves this by converting numerous patient stories into meaningful information. Patient reviews, surveys, EHRs, chatbots, call transcripts, and more can provide key insights into patient emotions and perceptions. While traditional satisfaction measures might simply lower these experiences to one number, sentiment analytics can detect the positive, negative, and more nuanced emotional reactions. Another layer of detail is added to the sentiment aspect by aspect-based analysis, which connects sentiment to specific aspects of the delivery of healthcare [Nigatu, 2024]. Positive feelers about nursing care may be accompanied by negative feelers about waiting time, communication, and/or environment at the facility. This is significant because it's critical to focus on the root cause of the dissatisfaction rather than taking an average satisfaction score as a measure of the patient's overall experience. This is where scheduling system integration and EHR interoperability come into play. Scheduling decisions may become unreliable if changes in the availability of a provider in the provider's system are not updated in other systems or if the provider's status as available or unavailable is not consistent [Umar, 2022]. Real-time and two-way information sharing can help minimize scheduling conflicts and help paint a more precise picture of available capacity. Self-scheduling, customisable reminders, and communication preferences based on

patient preference are all ways of enhancing patient engagement. Sentiment analytics can help to identify what patients are feeling, appointment analytics can help to organise appointments and resources, and queueing models can help to understand the influence of demand and capacity on the waiting and congestion experienced [Hur, 2024]. Combining these sources can yield insights that can be used to guide operational priorities, and operational data can be used to understand patient experiences. For instance, if frequent complaints have been made about long waits, then this can be considered in conjunction with arrival trends, staffing, appointment density, and length of service. This will then inform the design and evaluation of an intervention which can be carried out and evaluated with operational indicators, and then with patient feedback. This integrated analytical environment is dependent upon data quality [Wang, 2020]. The information may come from different and heterogeneous sources that are not always complete and reliable, and do not always use the same terminology or structure. If not standardized and validated, data collection and integration errors can impact sentiment scores, scheduling decisions, and queueing models. Therefore, it is crucial to implement measures like deduplication, format standardization, validation procedures, ongoing monitoring, and employee training. Good data governance also facilitates the ability to connect patient experiences with operational and clinical outcomes. The final analytical result is always dependent upon the input to the system. There are more methodological problems with cross-source linking. Different patients or events can be mentioned in different ways in patient reviews, clinical notes, EHR, scheduling data, and communication transcripts [Goyal, 2021]. To ensure meaningful relationships without duplication or inaccuracies, there needs to be a set of standardized schemas and the linking mechanisms must be appropriate. Meanwhile, historical or benchmark-based models may not apply to new populations, organizations, and/or time periods. External validation should, therefore, be integrated into the framework to evaluate the potential for analytical performance to be consistent across environments [Salman, 2020]. Continuous monitoring is of special significance, as healthcare data and processes will change over time. There's more to implementation in healthcare organizations than just a technical deployment. System design and system evaluation should involve organizational leadership, frontline staff, patients, data specialists, clinicians, and operational managers. Staff involvement is especially critical as health care employees know the "how" of the process and can detect issues that could not be seen in the data. Data quality, privacy, understanding the analytical outputs, recognition of errors, and patient feedback to be used should be taught at the training [Gross, 2018]. A non-punitive organizational culture can encourage employees to participate in quality improvement processes and patients to honestly report back. In general, the framework indicates that maximizing data use to improve patient outcomes and optimize operations is best achieved by taking into account both patient experiences and operational performance. Sentiment analytics allows one to understand the patients' perception, intelligent scheduling helps coordinate demand and resources, and queueing theory helps to gain quantitative insight into the flow and capacity of patients [Mazumder, 2025]. They rely heavily on the quality of the data, the ability of the data to be interoperable, the validation of the data, the protection of privacy, ethical governance, and human judgment. These elements can be combined to form a continuous improvement system in which patient experiences turn into operational knowledge and operational changes are assessed by patient-centered outcomes that are measurable.

7. Conclusion

Data-driven patient-centered healthcare improvement can be strengthened through the integration of sentiment analytics, intelligent scheduling, and queueing theory. Sentiment analytics captures patient experiences, while scheduling and queueing approaches improve resource allocation, patient flow, and waiting time management. Reliable data quality, interoperability, privacy protection, ethical governance, and continuous validation are essential for responsible implementation. Human oversight is also necessary to address bias, contextual limitations, and fairness concerns. Overall, integrating patient feedback with operational analytics can support continuous quality improvement, better access, reduced delays, improved resource utilization, and more responsive healthcare services while maintaining patient trust, safety, and equity.

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References

- [1] Adhicandra I., Nurhidayati S., and Fauzan T. R., (2024) Optimization of hospital queue management using priority queue algorithm and reinforcement learning for emergency service prioritization, *Int. J. Softw. Eng. Comput. Sci. (IJSECS)*, vol. 4, pp. 512–522, 2024. doi: 10.35870/ijsecs.v4i2.2772.
- [2] Akinode J. L. and Oloruntoba S. A., (2017) Design and implementation of a patient appointment and scheduling system, *Int. Adv. Res. J. Sci. Eng. Technol.*, vol. 4, pp. 16–23, 2017. doi: 10.17148/IARJSET.2017.41203.
- [3] Alam M. N., Sallem A., Pereira P., Bachir B., and Masmoudi N., (2023) Optimal artificial neural network using particle swarm optimization, *E3S Web Conf.*, vol. 469, p. 00019, 2023. doi: 10.1051/e3sconf/202346900019.
- [4] AISerkal Y. M. *et al.*, (2025) Real-time analytics and AI for managing no-show appointments in primary health care in the United Arab Emirates: Before-and-after study, *JMIR Form. Res.*, vol. 9, p. e64936, 2025. doi: 10.2196/64936.
- [5] Ang K. M. *et al.*, (2022) A modified particle swarm optimization algorithm for optimizing artificial neural network in classification tasks, *Processes*, vol. 10, no. 12, p. 2579, 2022. doi: 10.3390/pr10122579.
- [6] Caldwell K. E., Kushner B. S., Donald C., and Wise P. E., (2024) Implementation of flexible days off improves surgical resident attendance of personal health appointments, perceived wellbeing, and sense of control, *J. Surg. Educ.*, vol. 81, pp. 1522–1528, 2024. doi: 10.1016/j.jsurg.2024.07.024.
- [7] Gao Y., Zhang Q., Lau C. K., and Ram B., (2022) Robust appointment scheduling in healthcare, *Mathematics*, vol. 10, no. 22, p. 4317, 2022. doi: 10.3390/math10224317.
- [8] Goyal A., Kaushik S., and Khan R., (2021) IoT-based cloud network for smart health care using optimization algorithm, *Inform. Med. Unlock.*, vol. 27, p. 100792, 2021. doi: 10.1016/j.imu.2021.100792.
- [9] Gross D., Shortle J. F., Thompson J. M., and Harris C. M., (2018) *Fundamentals of Queueing Theory*, 5th ed. Hoboken, NJ, USA: John Wiley & Sons, 2018. doi: 10.1002/9781119453765.
- [10] Haleem A., Javaid M., Singh R. P., and Suman R., (2021) Telemedicine for healthcare: Capabilities, features, barriers, and applications, *Sens. Int.*, vol. 2, p. 100117, 2021. doi: 10.1016/j.sintl.2021.100117.
- [11] Han Y., Johnson M. E., Shan X., and Khasawneh M., (2024) A multi-appointment patient scheduling system with machine learning and optimization, *Decis. Anal. J.*, vol. 10, p. 100392, 2024. doi: 10.1016/j.dajour.2023.100392.
- [12] Hur M. R. and Al Akhir, (2026) Transforming healthcare quality through patient sentiment analytics: A data-driven framework for patient-centered quality improvement, *CAJMNS*, vol. 7, no. 2, pp. 677–686, Apr. 2026. [Online]. Available: <https://cajmns.casjournal.org/index.php/CAJMNS/article/view/3346>
- [13] Hur M. R., (2024) Intelligent scheduling systems and healthcare workforce efficiency: A survey-based study of 155 U.S. healthcare professionals, *Business and Social Sciences*, vol. 2, no. 1, pp. 1–9, 2024, Art. no. 10874. doi: 10.25163/business.2110874.
- [14] Kammrath B P. *et al.*, (2025) Efficient patient care in the digital age: Impact of online appointment scheduling in a medical practice and a university hospital on the 'no-show ' rate, *Front. Digit. Health*, vol. 7, p. 1345678, 2025. doi: 10.3389/fdgth.2025.1567397.
- [15] Knight R. T. D. *et al.*, (2023) Artificial intelligence for patient scheduling in the real-world health care setting: A metanarrative review, *Health Policy Technol.*, vol. 12, p. 100824, 2023. doi: 10.1016/j.hlpt.2023.100824.
- [16] Laan C., van de Vrugt M., Olsman J., and Boucherie R. J., (2018) Static and dynamic appointment scheduling to improve patient access time, *Health Syst.*, vol. 7, pp. 148–159, 2018. doi: 10.1080/20476965.2017.1403675.
- [17] Li N., Li N., Zhang C., and Kong N., (2021) Integrated optimization of appointment allocation and access prioritization in patient-centred outpatient scheduling, *Comput. Ind. Eng.*, vol. 154, p. 107125, 2021. doi: 10.1016/j.cie.2021.107125.
- [18] Mazumder M. S. A. and Hur M. R., (2025) Bridging queueing theory and data-driven analytics to reduce patient wait times: A cross-sectional survey of healthcare users in the United States, *Journal of Primeasia*, vol. 6, no. 1, pp. 1–11, 2025, Art. no. 10837. doi: 10.25163/primeasia.6110837.
- [19] Nigatu D. T., Dinka T. G., and Tilahun S. L., (2024) Convergence analysis of particle swarm optimization algorithms for different constriction factors, *Front. Appl. Math. Stat.*, vol. 10, p. 1304268, 2024. doi: 10.3389/fams.2024.1304268.
- [20] Niu T. *et al.*, (2024) A review of optimization studies for system appointment scheduling, *Axioms*, vol. 13, no. 1, p. 16, 2024. doi: 10.3390/axioms13010016.

- [21] Obermeier G., Zimmermann R., and Auinger A., (2020) The effect of queuing technology on customer experience in physical retail environments, in *HCI in Business, Government and Organizations*, F. F.-H. Nah and C.-H. Tan, Eds. Cham, Switzerland: Springer, 2020, vol. 12204, pp. 141–157. doi: 10.1007/978-3-030-50341-3_12.
- [22] Olayode I. O., Tartibu L. K., Okwu M. O., and Ukaegbu U. F., (2021) Development of a hybrid artificial neural network-particle swarm optimization model for the modelling of traffic flow of vehicles at signalized road intersections, *Appl. Sci.*, vol. 11, no. 18, p. 8387, 2021. doi: 10.3390/app11188387.
- [23] Rahman M. S., Islam S., and Ridoy M. J. I., (2026) Perceptions of AI-enabled public health among healthcare professionals: A cross-sectional survey on telemedicine access, disease surveillance, and operational efficiency, *J. AI ML DL*, vol. 2, no. 1, pp. 1–11, 2026, doi: 10.25163/ai.2110797.
- [24] Ridoy M. J. I., (2022) Predicting supply chain resilience with machine learning: A comparative analysis of ensemble and neural approaches across logistics, manufacturing, healthcare, and agriculture, *J. Primeasia*, vol. 3, no. 1, pp. 1–13, 2022, doi: 10.25163/primeasia.3110776.
- [25] Saad G., Harb H., Abouaissa A., Idoumghar L., and Charara N., (2023) A sensing-based patient classification framework for efficient patient-nurse scheduling, *Sustain. Comput. Inform. Syst.*, vol. 38, p. 100855, 2023. doi: 10.1016/j.suscom.2023.100855.
- [26] Salman O. H., Aal-Nouman M. I., and Taha Z. K., (2020) Reducing waiting time for remote patients in telemedicine with considering treated patients in emergency department based on body sensors technologies and hybrid computational algorithms: Toward scalable and efficient real time healthcare monitoring system, *J. Biomed. Inform.*, vol. 112, p. 103592, 2020. doi: 10.1016/j.jbi.2020.103592.
- [27] Shama A. T. and Biswas A., (2023) Cloud Misconfiguration as a Governance Failure in AI-Enabled Healthcare and Finance with Privacy Risk Implications, *International Journal on Economics, Finance and Sustainable Development*, vol. 5, no. 1, pp. 208–214, 2023. doi: 10.31149/ijefsd.v5i1.5799.
- [28] Umar M. et al., (2022) Swarming computational approach for the heartbeat Van Der Pol nonlinear system, *Comput. Mater. Contin.*, vol. 72, pp. 6185–6202, 2022. doi: 10.32604/cmc.2022.027970.
- [29] Valenzuela-Núñez C., Latorre-Núñez G., and Troncoso-Espinosa F., (2024) Smart medical appointment scheduling: Optimization, machine learning, and overbooking to enhance resource utilization, *IEEE Access*, vol. 12, pp. 7551–7562, 2024. doi: 10.1109/ACCESS.2024.3349953.
- [30] Wang D. et al., (2020) Clinical characteristics of 138 hospitalized patients with 2019 novel coronavirus-infected pneumonia in Wuhan, China, *JAMA*, vol. 323, no. 11, pp. 1061–1069, 2020. doi: 10.1001/jama.2020.1585.
- [31] Yadav A. and Roy S. M., (2023) An artificial neural network-particle swarm optimization (ANN-PSO) approach to predict the aeration efficiency of venturi aeration system, *Smart Agric. Technol.*, vol. 4, p. 100230, 2023. doi: 10.1016/j.atech.2023.100230.
- [32] Yadav R. K., (2019) GA and PSO hybrid algorithm for ANN training with application in medical diagnosis, in *Proc. Third Int. Conf. Intelligent Computing in Data Sciences (ICDS)*, Marrakech, Morocco, 2019, pp. 1–5. doi: 10.1109/ICDS47004.2019.8942375.
- [33] Yang Y., Madanian S., and Parry D., (2024) Enhancing health equity by predicting missed appointments in health care: Machine learning study, *JMIR Med. Inform.*, vol. 12, p. e48273, 2024. doi: 10.2196/48273.
- [34] Yousif A. B., Hassan H. J., and Muttasher G., (2022) Applying reinforcement learning for random early detection algorithm in adaptive queue management systems, *Indones. J. Electr. Eng. Comput. Sci.*, vol. 26, no. 3, pp. 1684–1691, 2022. doi: 10.11591/ijeecs.v26.i3.pp1684-1691.
- [35] Zhang H., Wang Y., Jia Y.-H., and Mei Y., (2026) Genetic programming with model-driven dimension repair for learning interpretable appointment scheduling rules, *IEEE Trans. Evol. Comput.*, early access, 2026. doi: 10.1109/TEVC.2026.3660890.
- [36] Zhou S., Li D., and Yin Y., (2021) Coordinated appointment scheduling with multiple providers and patient-and-physician matching cost in specialty care, *Omega*, vol. 101, p. 102285, 2021. doi: 10.1016/j.omega.2020.102285.
- [37] Zuo J., Jin Y., and Liu W., (2024) Outpatient scheduling problem in smart hospital with two-agent deep reinforcement learning algorithm, *Law. Probab. Risk*, vol. 27, pp. 45–67, 2024. doi: 10.1007/s10791-024-09474-1.